

Lecture 9 - Sep 28

Chapter 5 - Binomial Coefficients

Pascal's Identity - proved before:

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

Can construct Pascal's triangle using this identity:

$$\begin{array}{c} \binom{0}{0} \\ \binom{1}{0}, \binom{1}{1} \\ \binom{2}{0}, \binom{2}{1}, \binom{2}{2} \\ \vdots \\ \binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n} \end{array}$$

5.2 - The binomial Theorem



THEOREM

For positive integer n :

$$\begin{aligned} (x+y)^n &= \sum_{k=0}^n \binom{n}{k} x^k y^{n-k} = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k = \sum_{k=0}^n \binom{n}{n-k} x^{n-k} y^k \\ &= \binom{n}{0} y^n + \binom{n}{1} x y^{n-1} + \binom{n}{2} x^2 y^{n-2} + \dots + \binom{n}{n-1} x^{n-1} y + \binom{n}{n} x^n \end{aligned}$$

When going from left to right its *expansion*, when its going from right to left its *evaluating the sum*.

$$(x+y)^n = \binom{n}{0} y^n + \binom{n}{1} x y^{n-1} + \binom{n}{2} x^2 y^{n-2} + \dots + \binom{n}{n-1} x^{n-1} y + \binom{n}{n} x^n$$

→ expansion
← evaluating the sum

Proof

Scratch work

$$\text{Write } (x+y)^n = \underbrace{(x+y)(x+y)\dots(x+y)}_{n \text{ times}}$$

We completely expand this product, using the distributive law, and group like terms.

When expanding the right side, we know there must be 2^n in the final expansion. This is because for any term, for every $(x+y)$ either the term gets multiplied by x or by y .

Now consider the term

$$\boxed{??} x^k y^{n-k}$$

Here we know we must have "taken" (multiplied) by x k times, and the rest of the times $(n - k)$ it got multiplied by y .

Therefore, the number coefficient is the number of different ways get to this term, or in other words, the number of ways to pick k from n total elements.

Another way to think about it is to consider a string made of x and y only. Consider a string with exactly k copies of x :

$$\begin{array}{cccccccc} x & y & y & x & x & y & x & y & y & x & y & x & x \\ & & & & & & & & & & & & k = 6 \end{array}$$

We could say this string could represent one way to get $x^k y^{n-k}$, by taken x on the first $(x + y)$ term, then taking y on the second $(x + y)$ term, and so on. If we consider any string with exactly k copies of x , it represents one way to get $x^k y^{n-k}$ from the $(x + y)$ factors. Since each of these ways adds exactly one to the coefficient, the value of the coefficient becomes the number of ways to pick exactly k copies of x , which is $\binom{n}{k}$.

Remark: This argument assumes terms are being grouped by x , but same argument applies for y by just swapping both.

Proof

$$\text{Write } (x + y)^n = \underbrace{(x + y)(x + y) \dots (x + y)}_{n \text{ times}}$$

We completely expand this product, using the distributive law, and group like terms.

Since, for each factor $(x + y)$, we either pick x or we pick y . There are 2^n terms in total, and each can be arranged as $x^k y^{n-k}$ for $k = 0, 1, \dots, n$. We obtain the term $x^k y^{n-k}$ by choosing x in k of the n factors and by default, choosing y in $n - k$ of the remaining factors. There are $\binom{n}{k}$ ways to do so. Thus, $(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$. $\square$

Remarks:

1. $(1 + x)^n = \sum_{k=0}^n \binom{n}{k} x^k$
2. $\binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{n} = 2^n = \underbrace{(1 + 1)^n}_{\text{binomial theorem}}$
3. $(1 - x)^n = \sum_{k=0}^n (-1)^k \binom{n}{k} x^{n-k}$
4. $\binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \binom{n}{3} + \dots = (1 - 1)^n = 0$

A consequence of this is:

$$\binom{n}{0} + \binom{n}{2} + \dots = \binom{n}{1} + \binom{n}{3} + \dots = \frac{2^n}{2} = 2^{n-1}$$

Which states that the number of even subsets (left side) is equal in number to the number of odd subsets (right side)

Example 1

$$\sum_{k=0}^n 2^k \binom{n}{k} = 3^n$$

Proof

$$3^n = (1+2)^n = \sum_{k=0}^n \binom{n}{k} 2^k \quad \square$$

Example 2

Prove that

$$\sum_{k=1}^n k \binom{n}{k} = n 2^{n-1}$$

Proof

$$\text{Let } f(x) = (1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + \binom{n}{n}x^n$$

Differentiate all sides of the equation:

$$\begin{aligned} f'(x) &= n * (1+x)^{n-1} = 0 + \binom{n}{1} + 2\binom{n}{2}x + 3\binom{n}{3}x^2 + \cdots + n\binom{n}{n}x^{n-1} \\ &= \sum_{k=1}^n k \binom{n}{k} x^{k-1} \end{aligned}$$

$$\text{Let } x = 1. \text{ Then } n * 2^{n-1} = \sum_{k=1}^n k \binom{n}{k} \quad \square.$$

Note: Must expand out sum to account for the $k = 0$ term missing on the equation from the question to not lose points!

Example 3

Prove that

$$\sum_{k=1}^n k^2 \binom{n}{k} = n(n+1)2^{n-2}$$

Proof

Same first step as example 2:

$$n(1+x)^{n-1} = \sum_{k=1}^n k \binom{n}{k} x^k$$

After, multiply both sides by x :

$$n(1+x)^{n-1} * x = \sum_{k=1}^n k \binom{n}{k} x^{k+1}$$

Now differentiate both sides:

$$\underbrace{n(1+x)^{n-1}}_{2^{n-1}} + n(n-1) \underbrace{(1+x)^{n-2}}_{2^{n-2}} x = \sum_{k=1}^n k^2 \binom{n}{k} x^{k-1}$$

Let $x = 1$:

$$n2^{n-1} + n(n-1)2^{n-2} = n * 2^{n-2}(2 + (n-1)) = \sum_{k=1}^n k^2 \binom{n}{k}$$

Combinatorial proof

In mathematics, a *combinatorial proof* means the following:

- A proof by double counting e.g. $\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$
- a bijective proof e.g. $\binom{n}{k} = \binom{n}{n-k}$

Example 4

Use a combinatorial proof to prove that

$$k \binom{n}{k} = n \binom{n-1}{k-1}, k \geq 1$$

$$\iff \binom{n}{k} \binom{k}{1} = \binom{n}{1} \binom{n-1}{k-1}$$

Proof

Given n people, we choose k members, and one gets designated as the leader. We have $\binom{n}{1}$ ways to 1st select the chair and $\binom{n-1}{k-1}$ ways to select the remaining members. Alternatively, we have $\binom{n}{k}$ ways to select a k members, and then $\binom{k}{1}$ ways to select the leader from the k chosen members.

Therefore,

$$\binom{n}{1} \binom{n-1}{k-1} = \# \text{ of ways to form a group of } k \text{ members with a leader} = \binom{n}{k} \binom{k}{1} \quad \square.$$

Example 5

Give a combinatorial proof that $\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}, n \geq k \geq 0$.

$$\sum_{k=0}^n \binom{n}{k} \binom{n}{n-k} = \binom{n+n}{n}$$

Picking n elements total ($k + (n-k) = n$), but from 2 different piles.

of ways to pick exactly n people, that can be either boys or girls. So, can either pick k boys and $n - k$ girls, or pick any person by combining both n sized groups of boys and girls.

Example 6

Give a combinatorial proof that $\sum_{k=0}^r \binom{m}{r-k} \binom{n}{k} = \binom{m+n}{r}$.

Can deduce from example 5 by $m = n = r$